CHAPTER 16

Global health and surgery

Learning objectives

To define:
- The term global surgery

To describe:
- The role of a global surgeon

To understand:
- That surgery can be cost-effective
- Concepts of surgery and impoverishment
- How essential surgery can be delivered through surgical healthcare delivery platforms

To appreciate:
- The importance of access to surgical care
- The global surgical workforce
- The importance of global surgical metrics and research

INTRODUCTION AND DEFINITION

Global health is the health of populations in the global context. Global surgery is surgery with an understanding of public health. Surgeons understand the needs of their individual patients, while public health adds the understanding of the surgical operations needed in the population. Global surgery aims to provide equitable and improved surgical care across the world.

Global surgeons are not system-specific surgeons but are 'specialised' in providing the surgical needs of their communities. A practising surgeon can be viewed as a 'retailer' for the individual patient, whereas a global surgeon is typically the 'wholesaler' of the surgical needs of the population. This means that all surgeons working in non-tertiary hospitals who perform life-saving and essential surgeries, guided by the prevalent burden of surgical disease, are global surgeons. There is a misconception that a global surgeon is primarily a high-income country (HIC) surgeon helping low- and middle-income country (LMIC) surgeons periodically or through surgical missions. The HIC surgeons and subspecialist surgeons will be able to mitigate only a very small part of the vast unmet need of surgical disease burden in LMICs. Global surgery focuses on improving surgical health systems in parts of the world with a high surgical burden of disease by the local surgeons in those communities.

SURGERY AS AN ESSENTIAL AND COST-EFFECTIVE INTERVENTION

With the decline in the burden of communicable diseases in the world, one-third of the total disease burden is now due to surgical disease, with the majority being injury and cancers. In 2015, responding to this epidemiological transition, the World Health Organization (WHO) declared surgery to be a part of public health at the World Health Assembly, a meeting of all health ministers. Previously, surgical and anaesthesia care were perceived as too expensive and too complex to be a public health priority in resource-poor settings. The fact that early surgery saves lives and boosts the economy encouraged health planners to put surgery in the essential group of services in any national health strategy. Since then, scaling up surgical and anaesthesia care became a new worldwide movement under the umbrella term of 'global surgery'.

The Disease Control Priorities group of the World Bank has clearly identified surgical procedures that address the substantial needs of populations. In the absence of investments in surgical care, case-fatality rates and lifetime costs to the individual and society are high for common and easily treatable conditions, including appendicitis, hernia, fractures, obstructed labour, congenital anomalies and breast and cervical cancer. Furthermore, the cost-effectiveness of surgery for cataracts, hydrocephalus, limb deformity, general surgery and cleft lip or palate repair is comparable to widely used public health strategies, such as the bacille Calmette–Guérin (BCG) vaccine, and is much greater than other standard public health measures, such as antiretroviral therapy. Surgery becomes more cost-effective when not delivered as isolated interventions, but rather as a group of interventions within a platform of clinical care, such as a district hospital.

ACCESS TO SURGICAL CARE

The Lancet Commission on Global Surgery (2015) estimated that 5 billion out of the 7 billion people on the planet do not have access to surgery (see ***Summary box 16.2***). This is concerning, as access to timely life-saving essential surgery is a fundamental need for all. Although barriers to surgical access appear to be pronounced in LMICs, there are also disadvantaged and vulnerable populations in HICs. Access to

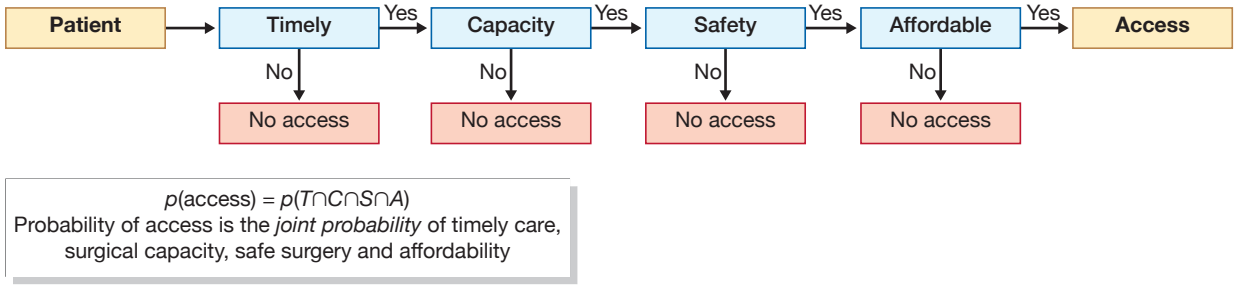

Figure 16.1 Access to surgery: the four dimensions.

surgery can be viewed through four lenses, namely; timeliness, capacity, safety and affordability (*Figure 16.1*).

Geographical access is the ability of a patient to reach a surgical facility within 2 hours, which is the crucial time for life-threatening haemorrhage. Capacity denotes that the facility has the required infrastructure and workforce and is able to perform safe surgery. However, the final barrier is when a patient cannot afford the surgery offered. A staggering nine out of 10 people in LMICs do not have access to surgery. Consequently, patients with acute surgical needs in LMICs do not reach hospital or reach it too late, in advanced stages of cancer, with already infected open fractures, with a perforated bowel or with burn contractures.

The world needs 143 million more surgical operations (unmet need) to be performed each year to save lives and prevent disability. Of the 313 million procedures undertaken worldwide each year, only 6% occur in the poorest countries, where over one-third of the world's population lives (see *Further reading*). Unmet need is greatest in eastern, western and central sub-Saharan Africa and South Asia. These regions have young mothers dying of physiological conditions such as pregnancy, for want of a caesarean section. The global surgical community hopes to achieve an optimum operative volume of 5000 surgical procedures per 100 000 population across the world by 2030. Currently, there are countries such as Ethiopia which do 150 operations per 100 000 population, while Hungary performs 23 000. However, more than 5000 operations and further expenditure do not bring commensurate health benefits to the population. Low operative volumes are also associated with high case-fatality rates from common, treatable surgical conditions, which include injuries, early cancer and burns.

THE GLOBAL SURGICAL WORKFORCE

A surgeon, anaesthetist and obstetrician (SAO) at the district hospital are considered essential staffing. In many LMICs the SAO density is less than 5 per 100 000 population. As the SAO number increases, there is a dramatic improvement in key indicators, such as the maternal mortality ratio. However, the benefits plateau beyond 20 SAOs per 100 000 population. By 2030, all LMICs are committed to scaling up their surgical workforce to at least 20 SAO providers per 100 000 population. For all countries to reach this benchmark 1.27 million providers will need to be trained by 2030. Where there is a workforce shortage with a high surgical burden of disease, 'task-shifting' is common, whereby appropriate tasks are moved to less specialised health workers. For example, essential life-saving anaesthesia is a short course for medical graduates to address the specialist anaesthetist shortage in rural areas. In countries where there are still too few medical graduates, task-shifting in both anaesthesia and surgery involves appropriately trained non-medically qualified clinicians or midwives.

Ironically, the countries in the world where the fewest children are born have the greatest number of paediatric surgeons. There is much redundancy in human resources and low volumes of surgery in many HICs. In countries with a large burden of surgical disease and large populations, often a single surgeon caters to the surgical needs of many millions of people. However, these global surgical champions have high burnout rates. Healthcare workers, including doctors and nurses, in LMIC settings are often mobile and may choose to migrate overseas, either staying in healthcare or taking up alternative professions (commonly referred to as the 'brain drain'). Twelve per cent of SAOs in HICs have graduated from LMICs and two-thirds of these graduates are from countries with fewer than 20 SAOs per 100 000 population. More surgeons and anaesthetists are needed in LMICs, but the trend cannot be mitigated by simply training more, as the migration rates and the in-country maldistribution of surgeons remain. The answer is starting global surgery units in teaching universities, e-grand rounds, remote specialist support, e-intensive care units, low-cost robotics, online learning platforms, supportive supervision, recognition, peer support and collegiality for upskilling these champions.

A surgical system goes beyond a surgeon and a sterile operating theatre. Clearly, it requires a contribution from many other people to make the surgical ecosystem work, both before and after the patient's visit to the operating theatre. Global surgery includes healthcare workers beyond anaesthetists, obstetricians and all surgical specialities. It encourages a multidisciplinary team approach to the profession and acknowledges the important role of physicians, nurses, public health practitioners, health managers, surgical, laboratory, supply-chain specialists, biomedical engineers, radiology and blood bank technicians. The role of community health workers in reducing delays and facilitating the referral pathway is vital in LMICs. Engaged and caring hospital managers are critical

in coordinating this process. Finally, private practitioners constitute a significant portion of the surgical manpower in LMICs and can contribute greatly to achieving the 2030 surgical burden goals, when appropriately incentivised and regulated for quality.

SURGERY AND IMPOVERISHMENT

Surgeons are tasked to operate and, in so doing, aim to successfully treat the surgical condition. A hernia not operated upon in a timely fashion costs the nation (and the individual) more when it becomes incarcerated. A delayed caesarean section puts mother and child at risk of death. In addition to risks associated with timing, the financial implications of a procedure can lead to poverty and catastrophic expenditure (more than 40% of annual household income). The one-time payment for surgery can push vulnerable populations into financial ruin, and those in poverty to extreme poverty. Forty-four per cent of the world's population is at risk of financial catastrophe with a single major operative procedure. In numbers, 33 million people (which is more than the population of Australia) each year will be pushed into poverty by paying for the surgery and anaesthesia that they need. Catastrophic expenditure is greatest for time-critical surgical conditions, such as peritonitis. Surgeons may have at times been less engaged with the patients' economic and social conditions that influence their health status. In many instances those who cannot afford to pay simply do not reach the operating theatre. Furthermore, lay people and policy makers alike can view surgery as an expensive business. This can lead to a problem, in that we can appreciate that everyone needs access to surgery, but commonly only the upper wealth quintiles are privileged to be in this position. When aggregated, the national estimate of out-of-pocket expenses of injury care alone exceeds the total health budget of many LMICs.

GLOBAL SURGICAL METRICS AND RESEARCH

Surgeons are familiar with vital-sign-based scoring systems for individual patients and with hospital metrics for inpatient hospital stay, surgical site infection or ventilator-associated pneumonia. Global surgery requires the addition of population-level metrics for the surgical burden of disease, which are less readily available. Household-level surveys of injury burden, vision care, cancer screening and worldwide metrics such as maternal mortality ratio and caesarean section rates are indicative of the surgical health systems in countries. Demographic surveillance systems and surveys such as the Million Death Study have been used to assess acute abdominal disorders at the national level. Most public health interventions are measured in terms of disability-adjusted life-years (DALYs) averted, which calculates how much it costs to avert 1 year of suffering due to a disability. One-third of all deaths worldwide are the result of conditions needing surgical care, and this surpasses human immunodeficiency virus, malaria and tuberculosis combined.

The core indicators to monitor the realisation of universal access to safe, affordable surgical and anaesthesia care are shown in *Table 16.1*.

TABLE 16.1 Core indicators to monitor the realisation of universal access to safe, affordable surgical and anaesthesia care.

Preparedness	a	Access to timely essential surgery (proportion of population within 2 hours of a facility that can perform the bellwether procedures)
	b	Density of surgeons, anaesthetists and obstetricians working per 100 000 population
Surgical service delivery	a	Procedures done in an operating theatre, per 100 000 population per year
	b	All-cause death rate before discharge of patients who have undergone a procedure in an operating theatre, divided by the total number of procedures
Affordability of surgery		Proportion of households protected against impoverishment and catastrophic expenditure from direct out-of-pocket payments for surgical care

ESSENTIAL SURGERY THROUGH SURGICAL HEALTHCARE DELIVERY PLATFORMS

The 44 essential surgeries listed by WHO are critical to life, and 29 of them can be done at a district hospital. The bellwether procedures include caesarean sections, laparotomies and treatment of open fractures. These serve as a proxy measure to gauge the functionality of the surgical health system and its ability to perform a broad range of other essential surgical procedures. In places where there are few specialist surgeons, surgical needs are triaged as: 'must-do' procedures (cannot wait for 24 hours), 'should-do' procedures (cannot be delayed beyond a week) and 'can-do' procedures (can wait for more than a week), rather than by specialty.

The suggested core packages for strengthening emergency and essential surgical care and anaesthesia as a component of universal health coverage are shown in *Summary box 16.1*.

National surgical plans consider country-specific contexts of disease burden, severity of disease, effectiveness of surgical intervention, economic effects and social implications. These plans influence decisions to tailor these procedures, packages and platforms for delivery. National standard treatment guidelines, which are commonplace in HICs, are now being adapted and used in the context of LMICs. They ensure that incentives for hospital management and clinical leadership align with the goal of efficient, system-wide reductions in the burden of surgical disease.

In the interconnected world, HICs can contribute in many important ways to global surgery: in elective and planned surgeries, academic grand rounds, relevant LMIC research

Summary box 16.1

Core packages for strengthening emergency and essential surgical care and anaesthesia

- *Emergency procedures packages* include:
  - Basic trauma package (e.g. fracture treatment, trauma laparotomy, debridement)
  - Basic obstetric package (e.g. caesarean section)
  - Basic emergency general surgical package (e.g. laparotomy, incision and drainage)
- *Planned care packages* can include:
  - General surgical package (e.g. hernia repair, bowel resection)
  - Obstetric and gynaecological package (e.g. hysterectomy)
  - Specialist surgical package (e.g. cataract, clubfoot correction)
  - Palliative surgical package (e.g. diversion colostomy, analgesics)

and research into the burden of surgical disease. Traditionally, mission surgeries for rarer conditions such as cleft lip and palate have contributed to high-quality protocols of safety and standards in surgery (Operation Smile, Smile Train). The critiques and concerns have been around HIC surgeons parachuting into LMICs and a lack of follow-up of their operated patients. Short-term visiting teams can draw resources away from local providers delivering continuous care, creating a perception within the community that visiting teams provide higher quality care. Improved LMIC institutional partnerships, local capacity building and collegiality are key to global surgery collaborative work with HICs. On their part, HICs have learned from LMIC institutions, such as the Aravind Eye Institute, about affordable surgery through remarkable cost reductions in high-volume cataract surgeries. While achieving high-volume surgeries, surgical safety is paramount and is dependent on training and upskilling of human resources in health, health infrastructure, the supply chain and equipment maintenance. Improved connectivity, infrastructure, Internet and wearables as well as low-cost simulation and robotics are remarkable global innovations that may make surgery accessible to those who were previously unable to reach it.

In the future, global surgery will drive surgeons in academic university hospitals to partner with their public health and community medicine colleagues for creating hub-and-spoke models for outreach surgery. Surgical practices will evolve to address the high unmet surgical burden of disease with high-volume and low-profit operations, with enhanced recovery in hospitals and postdischarge follow-ups by community health workers. Surgeons will go beyond the technical aspects of surgical practice to advocate for affordable and equitable surgical care for everyone, without compromising on safety.

Summary box 16.2

Key messages from the Lancet Commission on Global Surgery

- 5 billion out of the 7 billion people on the planet cannot access the surgeons who read this book for safe and affordable surgery
- 143 million more surgical procedures are needed each year in the world
- 33 million people each year will be impoverished because of paying for the surgery and anaesthesia that they need
- Investing in surgery is affordable, saves lives and promotes economic growth
- Surgery is an indivisible, indispensable part of health care. Surgical and anaesthesia care should be an integral component of a national health system in countries at all levels of development

FURTHER READING

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